SkillGauge AI – Interview Assessment Report

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1. You are building a web application using Flask to serve machine learning predictions. Which component is primarily responsible for handling incoming HTTP requests and routing them to the correct prediction function?
A) A Flask route decorator
B) A MongoDB query
C) A BigQuery stored procedure
D) A CSS stylesheet
Student Answer: C
Correct Answer: A
Marks: 0/1

2. You have a large dataset stored in BigQuery that needs to be processed for a machine learning model. Which Python library is most commonly used for efficient interaction with BigQuery?
A) pandas-gbq
B) psycopg2
C) pymongo
D) BeautifulSoup
Student Answer: D
Correct Answer: A
Marks: 0/1

3. When developing a complex ML pipeline in Dataiku, what is the primary benefit of using visual flow charts and recipes?
A) To enforce strict adherence to a predefined set of C algorithms
B) To enable rapid iteration, collaboration, and understanding of data transformations
C) To automatically generate all necessary HTML and CSS for a user interface
D) To directly integrate with AR/VR rendering engines
Student Answer: B
Correct Answer: B
Marks: 1/1

4. You are encountering performance issues with a computationally intensive machine learning model training process. Which of the following cloud services on Google Cloud is best suited for accelerating this task?
A) Cloud Run
B) Compute Engine with GPUs/TPUs
C) Cloud Storage
D) App Engine
Student Answer: C
Correct Answer: B
Marks: 0/1

5. In a C program designed to process image data for an AI application, what is the typical memory management concern when dealing with large image buffers?
A) Ensuring proper garbage collection of unused variables
B) Preventing memory leaks by correctly allocating and deallocating memory
C) Managing the lifecycle of HTML DOM elements
D) Optimizing SQL query execution plans
Student Answer: D
Correct Answer: B
Marks: 0/1

6. You've made several changes to your Python codebase for a machine learning project and want to share them with your team. What is the primary function of Git in this workflow?

A) To automatically compile your C code into an executable
B) To manage and track changes to your codebase, enabling collaboration
C) To style your HTML and CSS for a responsive user interface
D) To visualize complex datasets in Spotfire
Student Answer: B
Correct Answer: B
Marks: 1/1

7. You are designing a user interface for a VR application that displays dynamic data. Which of the following technologies is LEAST relevant for rendering the 3D environment and handling user interactions within VR?
A) Unity or Unreal Engine
B) HTML and CSS
C) WebXR API
D) Three.js
Student Answer: B
Correct Answer: B
Marks: 1/1

8. When training a machine learning model, what is the primary purpose of splitting your dataset into training, validation, and test sets?
A) To increase the overall size of the training data
B) To evaluate the model's generalization performance and prevent overfitting
C) To speed up the model training process
D) To directly embed the model into a SQL database
Student Answer: C
Correct Answer: B
Marks: 0/1

9. You are using VS Code to develop a Flask application. Which feature of VS Code is most useful for debugging runtime errors in your Python code?
A) The integrated terminal
B) The debugger with breakpoints and variable inspection
C) The Git integration
D) The CSS IntelliSense
Student Answer: C
Correct Answer: B
Marks: 0/1

10. You need to store user profile information for a web application. Which type of database is generally more flexible for handling semi-structured or rapidly evolving data schemas compared to a relational database like SQL?
A) MongoDB
B) BigQuery
C) PostgreSQL
D) SQLite
Student Answer: C
Correct Answer: A
Marks: 0/1

11. In the context of ML pipeline development, what does 'model training' primarily involve?
A) Designing the user interface using HTML and CSS
B) Adjusting model parameters based on input data to learn patterns
C) Setting up a cloud infrastructure on Google Cloud
D) Writing C code to optimize database queries
Student Answer: C
Correct Answer: B
Marks: 0/1

12. You are building a dashboard in Spotfire to visualize the results of a machine learning model. Which type of data source is Spotfire typically best suited for connecting to for interactive analysis?
A) Raw, unstructured text files
B) Structured or semi-structured data from databases or files
C) Real-time AR/VR sensor streams
D) Compiled C executables
Student Answer: D
Correct Answer: B
Marks: 0/1

13. You are tasked with creating a responsive web page. Which CSS property is crucial for controlling the layout and alignment of elements on different screen sizes?
A) color
B) font-size
C) display: flex / grid
D) background-image
Student Answer: C
Correct Answer: C
Marks: 1/1

14. Which of the following is a core concept in Artificial Intelligence that aims to mimic human cognitive abilities like learning and problem-solving?
A) Relational algebra
B) Object-oriented programming
C) Machine learning
D) Network protocols
Student Answer: D
Correct Answer: C
Marks: 0/1

15. You need to store and query large volumes of structured data for analytical purposes. Which Google Cloud service is specifically designed for this task?
A) Cloud Functions
B) BigQuery
C) Cloud Storage
D) Kubernetes Engine
Student Answer: D
Correct Answer: B
Marks: 0/1

16. You are writing a Python script to interact with a MongoDB database. What is the primary purpose of a 'collection' in MongoDB?
A) To define the schema for all documents
B) To group related documents together
C) To execute SQL queries
D) To manage Git repositories
Student Answer: C
Correct Answer: B
Marks: 0/1

17. When developing a machine learning model, what is the primary concern addressed by techniques like regularization (e.g., L1, L2)?
A) To increase the model's complexity
B) To reduce the model's training time
C) To prevent overfitting by penalizing large coefficients
D) To directly embed the model into an HTML page
Student Answer: C
Correct Answer: C

Marks: 1/1

18. You are collaborating on a C project using Git. After making changes, what is the command you would typically use to save those changes to your local repository?
A) git push
B) git pull
C) git commit
D) git clone
Student Answer: C
Correct Answer: C
Marks: 1/1

19. Which of the following describes the core difference between AR (Augmented Reality) and VR (Virtual Reality)?
A) AR overlays digital information onto the real world, while VR replaces the real world with a simulated one.
B) VR is used for gaming, while AR is used for business applications.
C) AR requires a headset, while VR does not.
D) VR is a subset of AR.
Student Answer: C
Correct Answer: A
Marks: 0/1

20. You are creating a simple web page and need to link a CSS file to your HTML document. Where would you typically place the `link` tag?
A) Within the `body` tag
B) Within the `head` tag
C) As an attribute of the `body` tag
D) After the closing `/html` tag
Student Answer: C
Correct Answer: B
Marks: 0/1

21. When building an ML pipeline, what is the role of data preprocessing?
A) To deploy the trained model to production
B) To transform raw data into a format suitable for model training
C) To visualize the model's performance metrics
D) To write documentation in C
Student Answer: C
Correct Answer: B
Marks: 0/1

22. You are developing a Python-based REST API using Flask. What is the primary purpose of using a framework like Flask in this scenario?
A) To manage complex SQL database schemas
B) To provide structure and tools for building web applications and APIs
C) To perform low-level memory management in C
D) To render 3D graphics for AR/VR experiences
Student Answer: D
Correct Answer: B
Marks: 0/1

23. You have a large dataset in BigQuery that you need to analyze and visualize. Which tool, from the given options, is primarily designed for interactive data visualization and business intelligence?
A) VS Code
B) Spotfire
C) MongoDB
D) Flask

Student Answer: B
Correct Answer: B
Marks: 1/1

24. In the context of Google Cloud, which service is best suited for running containerized applications at scale, often used for deploying machine learning models or web services?
A) Cloud SQL
B) Cloud Functions
C) Google Kubernetes Engine (GKE)
D) Cloud Storage
Student Answer: B
Correct Answer: C
Marks: 0/1

25. You are working with C and encounter a situation where you need to efficiently search for specific patterns within a large text file. Which standard library function is commonly used for this purpose?
A) malloc()
B) printf()
C) strstr()
D) scanf()
Student Answer: A
Correct Answer: C
Marks: 0/1

FINAL SCORE: 7/25